



Toronto Robbery Analysis

Advanced Data Analytics Tools

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PROJECT OBJECTIVE



- Analyze robbery trends using Toronto Police public data
- Identify patterns (time, location, offence type)
- Develop data-driven prevention strategies
- Intended audience: **Toronto Police Service Crime Strategy Unit**



DATASET OVERVIEW

Source: **Toronto Police Service Robbery Open Data**

Variables:

-  Neighbourhood, Division
-  Hour, Day, Month, Season
-  Location type (street, store, transit)
-  Type of offence

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
OBJECTID	EVENT_UN	OCC_YEAR	OCC_MON	OCC_DAY	OCC_DOY	OCC_DOW	OCC_HOU	DIVISION	LOCATION	PREMISES	OFFENCE	MCI_CATE	HOOD_15	NEIGHBOU	LONG_WG	LAT_WGS	MonthCha	MonthNum	Month_Start	HourNum	TimeOfDay
346	GO-20141	2014	February	5	36	Wednesday	21:00	NSA	Convenience	Commercial	Robbery With	Robbery	NSA	NSA	0	0	FEBRUARY	2	01-02-2014	21	Evening
3	GO-20141	2014	January	1	1	Wednesday	02:00	NSA	Streets, Road	Outside	Robbery - In	Robbery	NSA	NSA	0	0	JANUARY	1	01-01-2014	2	Late Night
1	GO-20141	2014	January	1	1	Wednesday	04:00	D14	Streets, Road	Outside	Robbery With	Robbery	078	Kensington	-79.402	43.6476	JANUARY	1	01-01-2014	4	Late Night
2	GO-20141	2014	January	1	1	Wednesday	14:00	D12	Apartment	Apartment	Robbery - In	Robbery	113	Weston (1	-79.516	43.7002	JANUARY	1	01-01-2014	14	Afternoon
5	GO-20141	2014	January	1	1	Wednesday	03:00	D51	Other Comm	Commercial	Robbery - In	Robbery	167	Church-W	-79.384	43.6639	JANUARY	1	01-01-2014	3	Late Night
8	GO-20141	2013	December	31	365	Tuesday	12:00	D41	Commercial	Commercial	Robbery - In	Robbery	119	Wexford/M	-79.302	43.7708	DECEMBER	12	01-12-2013	12	Afternoon
10	GO-20141	2014	January	2	2	Thursday	10:00	D41	Parking Lot	Outside	Robbery With	Robbery	120	Clairlea-Bi	-79.299	43.7228	JANUARY	1	01-01-2014	10	Morning
7	GO-20141	2014	January	2	2	Thursday	19:00	D42	Parking Lot	Outside	Robbery - In	Robbery	146	Malvern Ea	-79.217	43.8043	JANUARY	1	01-01-2014	19	Evening
6	GO-20141	2014	January	2	2	Thursday	20:00	D33	Parking Lot	Outside	Robbery - In	Robbery	047	Don Valley	-79.344	43.7759	JANUARY	1	01-01-2014	20	Evening

DATA CLEANING (SAS) 🧹

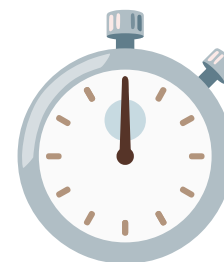
Removed duplicates & unnecessary fields



Fixed inconsistent formats (Street vs Roadway)

Standardized time & created:

- Time-of-Day Variable
- Holiday Flag
- Season Categorization



TOOLS USED



- Cleaned and prepared the raw robbery dataset
- Removed duplicates, fixed timestamps, and standardized labels
- Created new variables for time-of-day, seasons, and holidays



- Built dashboards, charts, and maps
- Visualized trends by hour, day, neighbourhood, and month
- Used forecasting tools to predict future robbery patterns



- Quick data checks after download
- Reviewed column names, missing values
- Used filters for basic quality checks before SAS

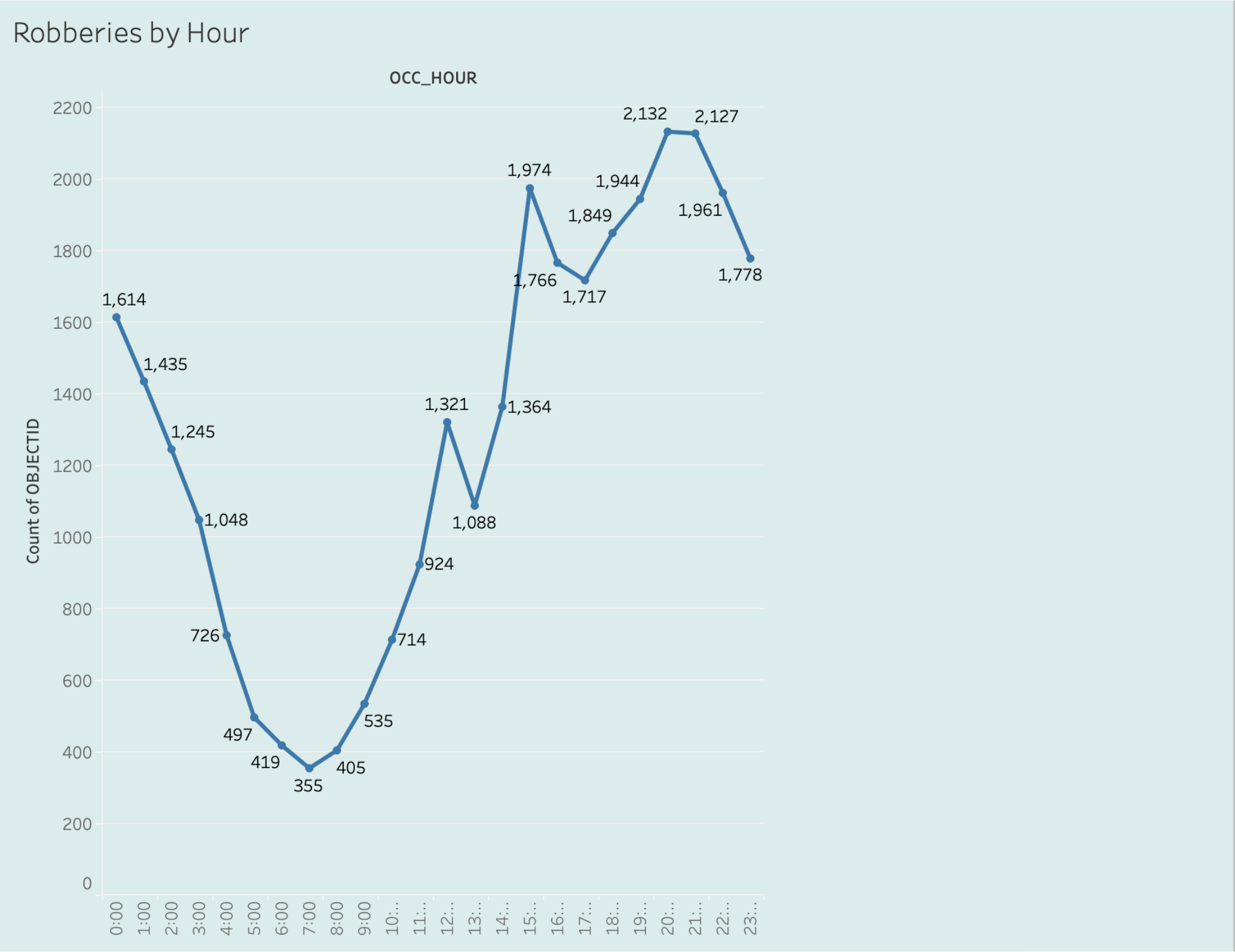
TIME TRENDS (BY HOUR)

Robberies peak during evening hours (6 PM — 11 PM)

Crime activity is lower in the early morning

Robberies increase sharply in the evening

Pattern follows after-work crowds, transit use, shopping, and nightlife



TIME TRENDS (BY DAY OF WEEK)

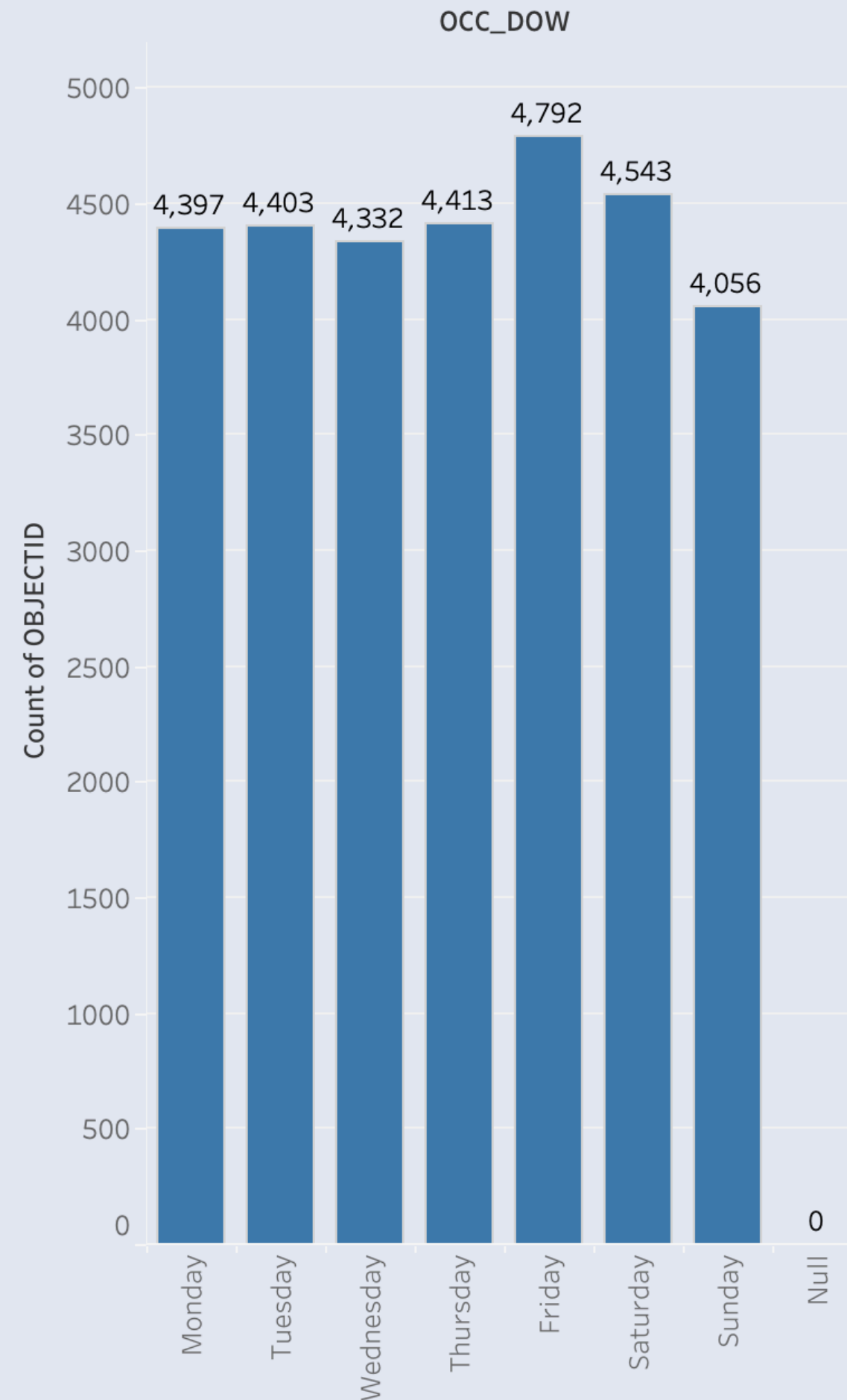
Robberies are slightly higher on Fridays & Saturdays

Weekday levels stay fairly consistent

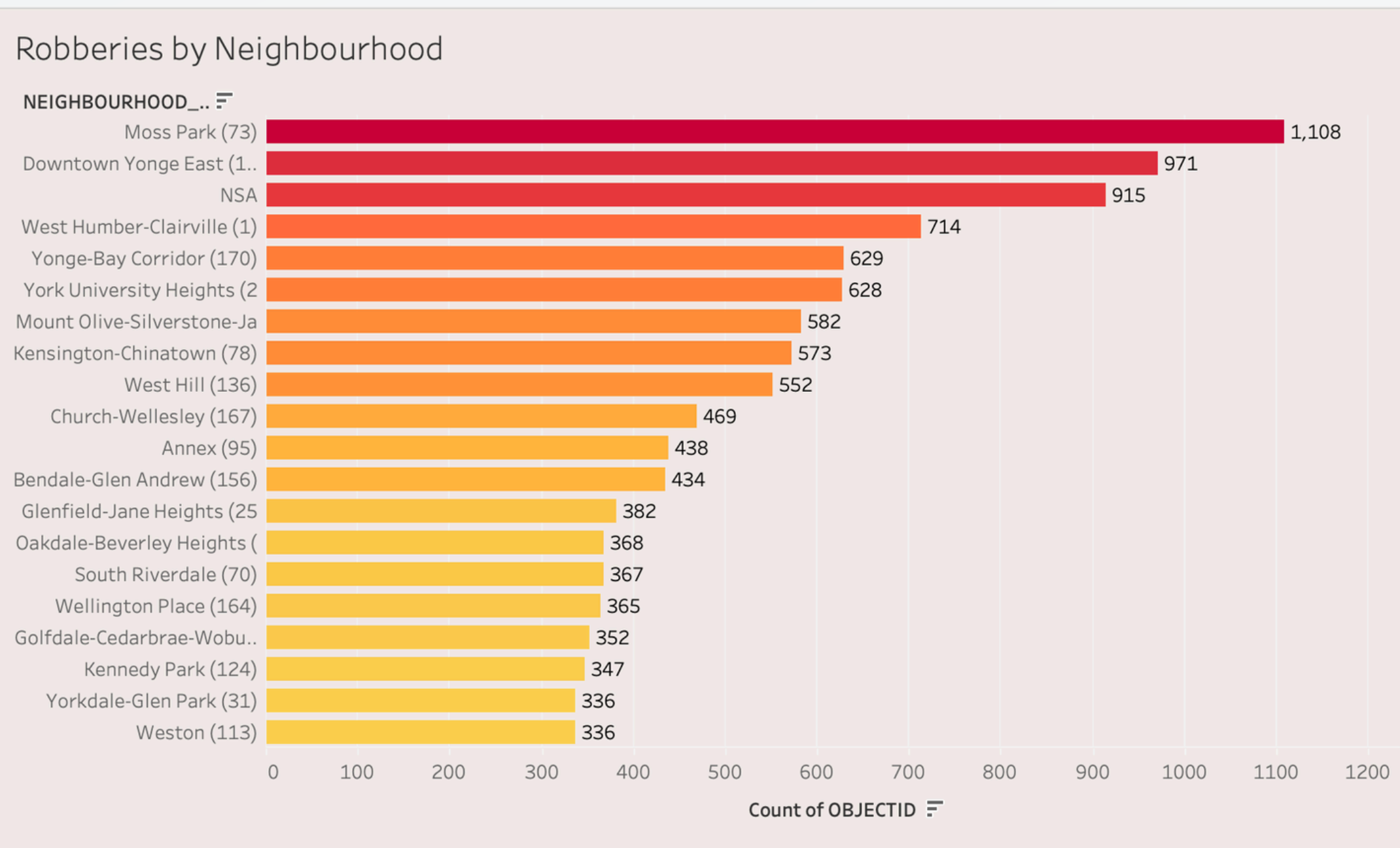
Helps identify when patrols should be increased

Increase likely due to weekend events, nightlife, and higher foot traffic

Robberies by Day of Week



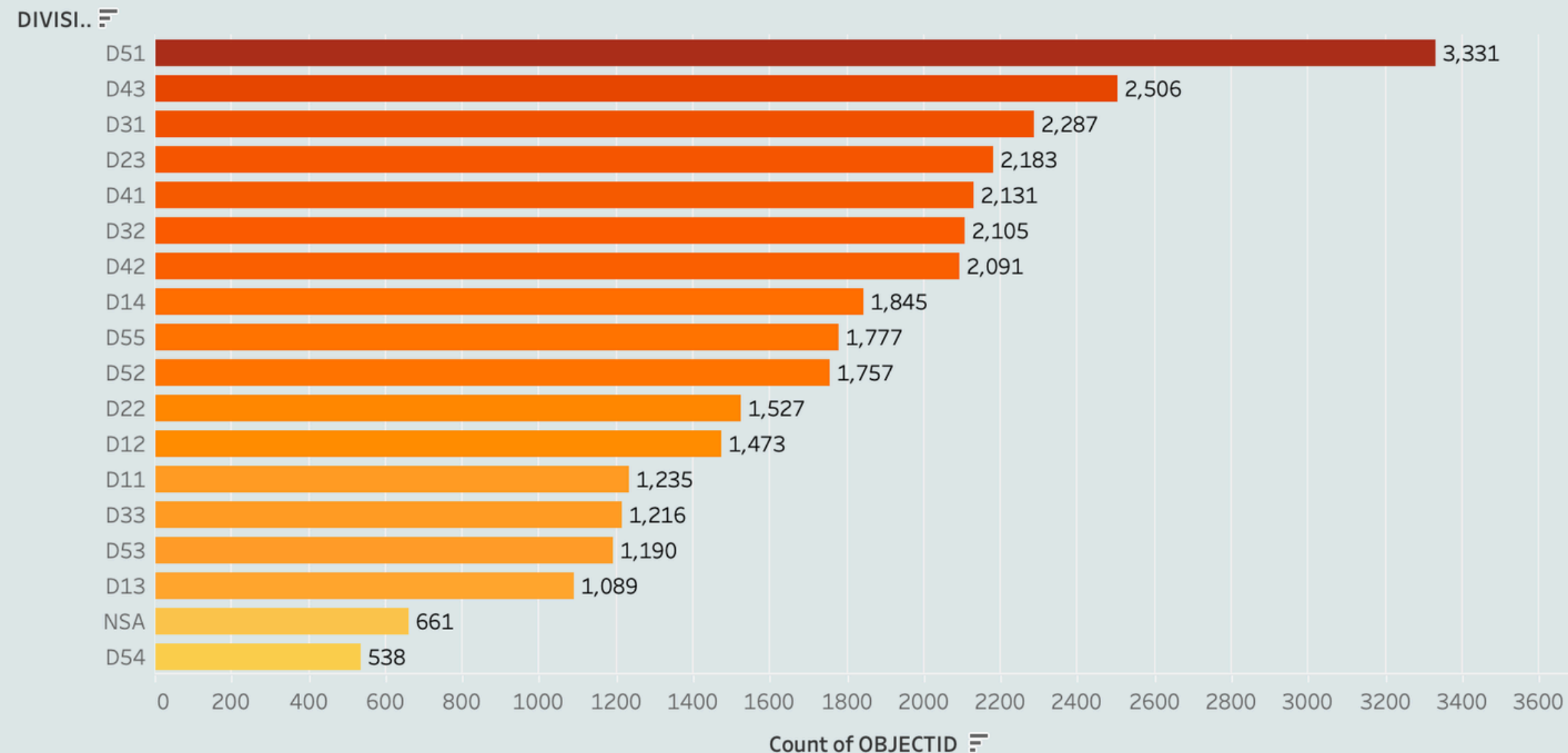
FINDS: LOCATION



- Moss Park has the highest robbery count (1,108 incidents)
- Followed by Downtown Yonge East and NSA
- Hotspots align with areas of high foot traffic, transit access, and commercial activity
- Confirms neighbourhood-level clustering of robbery incidents

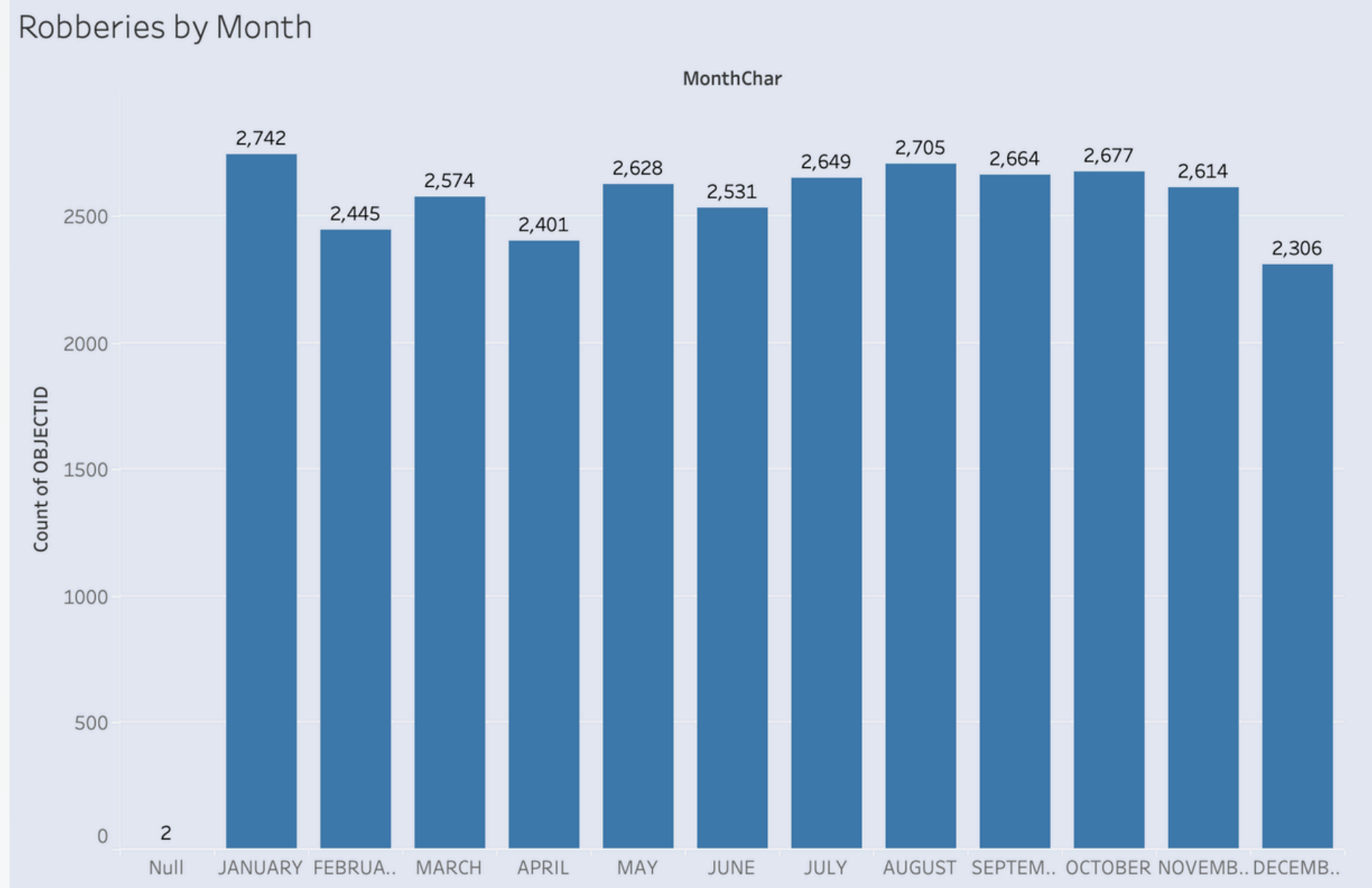
FINDS: LOCATION

Robberies by Division



- Division 51 has the highest robbery count (3,331 incidents)
- Followed by Divisions 43, 31, 23, and 41
- These divisions cover downtown, Scarborough transit corridor, and busy commercial zones
- Indicates strong clustering in high-density, high-mobility areas

FINDS: SEASONAL PATTERNS



- Higher robbery activity from May to October
- Peak months: July, August, October
- Lowest in January–March
- Seasonal trends reflect changes in outdoor activity and opportunity

FINDS: SEASONAL PATTERNS



- Robbery totals fluctuate but remain high
- Sharp drop in 2020–2021, followed by recovery
- Recent years show upward trend
- Forecast predicts 6–9% increase if conditions continue



RECOMMENDATIONS FOR IMPROVING SAFETY

- Increase police patrols during peak robbery hours (6 PM–11 PM)
- Focus extra resources on high-risk neighbourhoods and divisions
- Add more surveillance and safety programs in hotspot areas
- Plan police staffing based on seasonal patterns (summer peaks)
- Increase presence and awareness during holiday period (Dec–Jan)
- Work with local businesses to prevent commercial robberies
- Use data regularly to adjust strategies and monitor trends





CONCLUSION

- SAS & Tableau helped identify robbery patterns across Toronto
- Robbery locations and times show clear, predictable trends
- Data shows both time-based and neighbourhood-based risks
- Insights support better planning and resource allocation
- Data-driven strategies can lower incidents and improve safety
- Strong foundation for future policing and community programs

